



Builder Basics — The Impossible Statue (Extension)

Topic: Place + Move, On and Off · **Course:** Builder Basics · **Type:** Extension Activity · **Time:** about 45 minutes

✚ This is a bonus challenge, not a weekly lesson. Try it once you are comfortable placing blocks and moving the agent.

1 · Read the World 🔍

Open the **Impossible Statue** world. Someone left you a puzzle. Before you write any code, read the markers:

-  **Red blocks** — your **start** point and your **end** point. The agent begins on one and must finish on the other.
-  **Crying obsidian** — this is the shape you must **build**. Every crying obsidian spot is a block you need to place.
-  **Glass** — a **trap**. You must move through these spots but place **nothing** here. Leave them empty.

In your own words, what is the puzzle asking you to do?

Walk the path with your eyes from the red start block to the red end block. How many blocks should you place, and how many spots should you skip?

2 • The Big Idea: Place On / Place Off

The agent moves in a straight line, but it should **not** place a block on every step. On the obsidian spots, placing is **on**. On the glass spots, placing is **off**.

So your code is a mix: `move forward`, and sometimes `place block down`, and sometimes just `move forward` with no place.

```
place block down
move forward
place block down
move forward
move forward
place block down
```

Look at the code above. On which steps is placing ON? On which step is it OFF (a skipped spot)?

Why can't you just place a block on every single step?

3 • Plan Your Path

Before you build in the world, write the plan on paper first. Imagine the row of spots from start to end. Mark each spot with B (

Example: B B S B S S B

Write your row of B and S, in order from the red start to the red end:

Now turn that plan into agent commands. Remember: a B spot needs a place and a move; an S spot needs only a move.

Write your commands:

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.

4 • Spot the Bug 🐛

Each code block was meant to copy the statue, but it is broken. Read the goal, then rewrite it so it works. Then explain the fix.

Bug A – The pattern should be: build, skip, build. (B S B)

```
place block down
place block down
move forward
move forward
place block down
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B – The agent placed a block on a glass (skip) spot. The pattern should be: build, skip, skip, build. (B S S B)

```
place block down
move forward
place block down
move forward
place block down
move forward
place block down
```

Write the fixed code:

Why was it wrong? Why does your fix work?

5 · Build the Statue 🗿

Now go to the Impossible Statue world and run your code. Match the crying obsidian exactly: place where it says build, leave the glass spots empty, and finish on the red end block.

When it looks right, send a photo or video. Then explain what you did. Use these starters – write 4 to 6 sentences.

First, I read the markers and saw ...

My plan was ... (B and S)

I used the command ... to place, and I left the skip spots empty by ...

The hardest part was ...

To fix it, I ...

If I had more time, I would ...

First, I read the markers and saw ...

My plan was ... (B and S)

I used the command ... to place, and I left the skip spots empty by ...

The hardest part was ...

To fix it, I ...

If I had more time, I would ...

First, I read the markers and saw ...

My plan was ... (B and S)

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I used the command ... to place, and I left the skip spots empty by ...

The hardest part was ...

To fix it, I ...

If I had more time, I would ...

6 • Record Your Walkthrough

Take a video on your phone (or a parent's phone) and teach someone how you solved the puzzle. Try to use these words: **place, skip, move forward, start, end, pattern.**

1. Point to the red start and red end blocks.
2. Show the crying obsidian shape you copied.
3. Show one glass spot you skipped and explain why you placed nothing there.
4. Read your commands out loud and say which ones are "place on" and which are "skip".

Write what you will say in your video. Plan it here first – you can read from it while filming.